This thesis examines the persistent "scale-up gap" in the European Union's deep technology sector, as reflected in the continued dominance of incumbent firms and comparatively lower rates of firm turnover than those observed in the comparative economies of the United States and China. It argues that the divergence cannot be explained solely by differences in entrepreneurial capacity nor capital availability, but is instead rooted in the institutional logic of the prevailing EU governance.

Whereas the United States and China have developed innovation frameworks that tolerate, or actively internalize, high levels of fundamental uncertainty through, respectively, mission-oriented public investment and state-led strategic coordination. The European Union operates within a governance system shaped by supranational legal constraints, market neutrality principles, and redistributive objectives. This thesis investigates whether these institutional features systematically limit the EU's capacity to support the forms of risk-taking and market creation required for deep tech scale-up.

The study advances the argument that EU innovation policy is organized around a default logic of actuarial risk management rather than uncertainty absorption. This orientation arises from the Union's legal and political mandates to preserve competitive neutrality and ensure balanced regional development. The central question is whether this actuarial logic constrains the EU's ability to engage in the kind of directed, experimental, and failure-tolerant intervention associated with mission-led technological leadership.

By examining the design selection criteria, and governance structures of key EU innovation instruments the research evaluates whether the prioritization of market stability and risk mitigation operates as a structural barrier to deep tech scaling, even in policy frameworks explicitly framed as "mission-oriented".

RQ1: *How do the innovation governance systems of the United States, China, and the European Union differ in their treatment of fundamental uncertainty versus calculable risk in the financing and regulation of deep technology development?*

RQ3: *How is the concept of "mission-oriented" innovation translated into EU administrative instruments, and to what degree does this translation emphasize regulatory and financial de-risking over technological directionality and scale? What are the implications of this emphasis for deep tech commercialization and industrial leadership?*